



Semester One Examination, 2024

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section One: Calculator-free

If required by your examination administrator, please place your student identification label in this box

WA student number: In figures

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In words

Circle your teacher:

Mrs Alvaro

Mr Buckland

Ms Chua

Mrs Fraser-Jones

Miss K

Mr Koulianos

Mrs Lastari

Ms Lee

Mr Lei

Mr Luzuk

Mr Ng

Mr Safa

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	6	6	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
1	6	
2	9	
3	12	
4	11	
5	7	
6	7	
Section 1 Total	52	
Weighted ($\times 0.6731$)	35%	
Section 2 Weighted	65%	
Total	100%	

Section One: Calculator-free

35% (52 Marks)

This section has **six** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(6 marks)

(a) Evaluate $^{100}C_{98}$

(2 mark)

(b) Determine the value of m ($m \geq 0$) in each of the following equations.

(i)
$$\binom{m}{0} + \binom{m}{1} + \binom{m}{2} + \binom{m}{3} = 2^m$$
 (1 mark)

(ii)
$$\binom{15}{2m} - \binom{15}{m+3} = 0$$
 (3 marks)

Question 2

(9 marks)

(a) Solve the following equations.

(i) $x^2 = 49.$

(1 mark)

(ii) $2x^2 = 5x$

(1 mark)

(iii) $3x^2 + 11x - 4 = 0.$

(2 marks)

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- (b) Determine the coordinates of the turning point of the graph of $y = (x + 2)(x - 8)$.
(2 marks)

- (c) The equation of the circle with centre $(-2, 0)$ and radius 3 is $x^2 + y^2 + ax + b = 0$.
Determine the value of the constant a and the value of the constant b . (3 marks)

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Question 3

(12 marks)

100 high school students were asked what pets they have. The results are as follows.

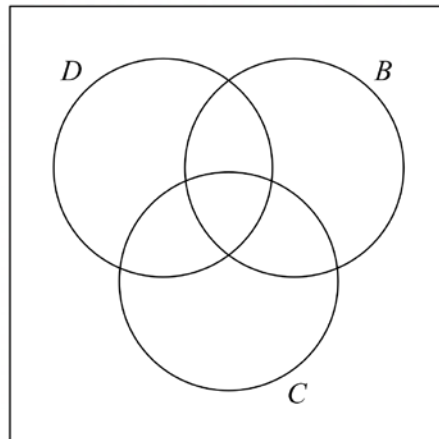
50 have a dog (D)
25 have a dog and birds

40 have birds (B)
14 have a dog and cat
7 have all three

46 have a cat (C)
12 have a cat and birds

- (a) Represent this information in the Venn diagram below.

(3 marks)



- (b) Describe the students in the set $(D \cup B) \cap C'$.

(1 mark)

- (c) Calculate the probability that a student has exactly two types of pets.

(1 mark)

- (d) Calculate the probability that a student has no pets.

(1 mark)

- (e) Calculate the probability that a student has a dog given that they have only one type of pet.
(2 marks)

One of the students claimed that owning a cat has no impact on whether you own a bird.

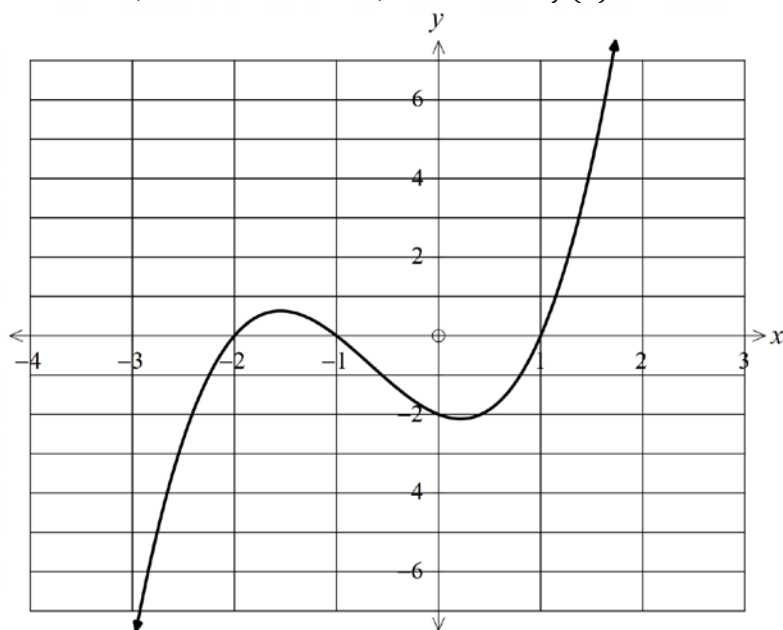
- (f) Comment on this claim, including suitable calculations to justify your response.
(4 marks)

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Question 4

(11 marks)

- (a) Determine, in factorised form, the function $f(x)$ shown on the graph below. (2 marks)



- (b) Expand the function $f(x)$ found in part (a).

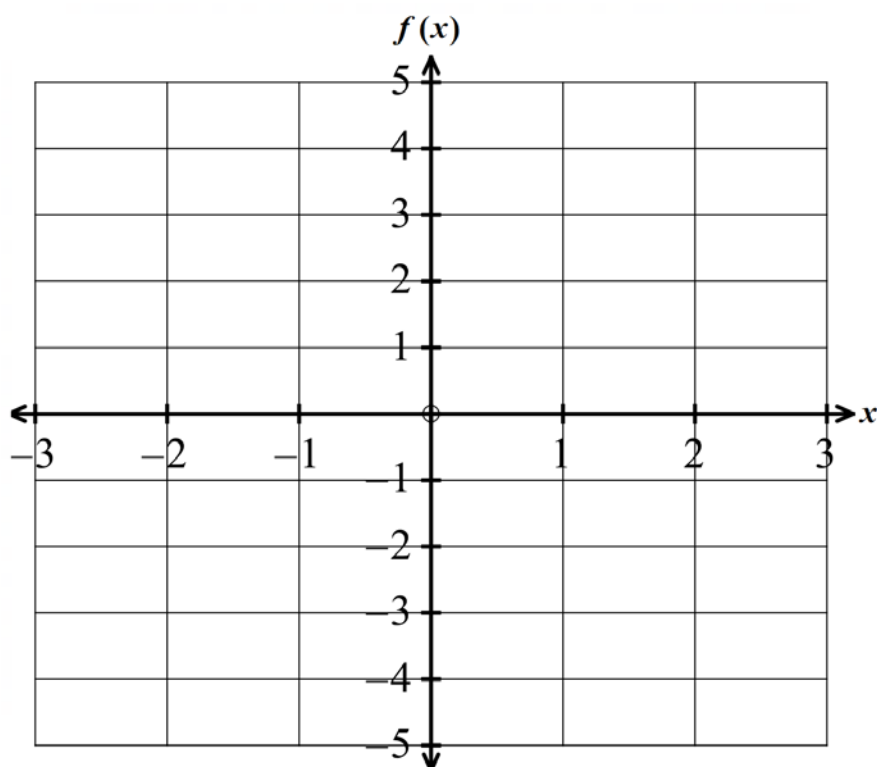
(2 marks)

A student is solving the following simultaneous equations:

$$y = x^2 + 2x \quad y = \frac{2}{x} + 1$$

- (c) Show that the student would need to solve $x^3 + 2x^2 - x - 2 = 0$. (2 marks)

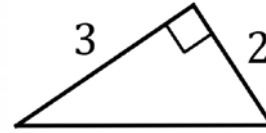
- (d) Sketch on the axes below the graphs of $y = x^2 + 2x$ and $y = \frac{2}{x} + 1$, clearly labelling all points of intersection. (5 marks)



Question 5

(7 marks)

- (a) Using the right triangle shown, or otherwise, if $\tan \lambda = 1.5$ and λ is an acute angle, determine the exact value of $\sin \lambda$.



(2 marks)

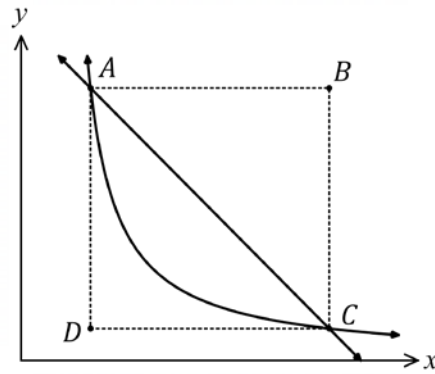
- (b) Determine all solutions of the equation $4 \cos(x - 11^\circ) - 2\sqrt{2} = 0$ for $0^\circ \leq x \leq 360^\circ$. (3 marks)

- (c) Determine the equation of the straight line that passes through the point $(0, -2)$ and has an angle of inclination of 30° with the positive x -axis. (2 marks)

Question 6

(7 marks)

The diagram, not drawn to scale, shows rectangle $ABCD$ with side AB parallel to the x -axis, so that A and C are the points of intersection of the line $y = 9 - x$ with the hyperbola $y = \frac{6}{x-1}$.



- (a) State the equations of the two asymptotes of the hyperbola. (2 marks)

- (b) Determine the area of rectangle $ABCD$. (5 marks)

End of questions

Supplementary page

Question number: _____

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